LAPORAN PRAKTIKUM

REKAYASA SISTEM INFORMASI

Backend: CRUD API

Disusun Oleh:

Trisha Garnis Wahningyun	(L0224012)
Alvian Damar Budhi Hernowo	(L0224014)
Fathul Fajar Nur Ikhsan	(L0224018)
Kunto Rossindu Hidayattullah	(L0224020)
Zaki Elias Al Haqqanikudus	(L0224039)

PROGRAM STUDI S1 SAINS DATA

FAKULTAS TEKNOLOGI INFORMASI DAN SAINS DATA

UNIVERSITAS SEBELAS MARET

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DOKUMENTASI PENGERJAAN

Tautan Repositori (Source Code)

Seluruh source code yang digunakan dalam penugasan implementasi operasi CRUD API ini, termasuk pengaturan arsitektur Controller, Service, Repository, dan Database Migration menggunakan Alembic, telah diunggah dan dapat diakses secara publik melalui repositori GitHub berikut:

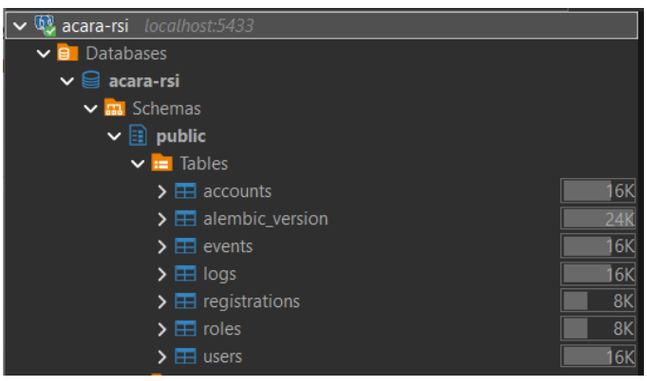
🔗 https://github.com/trishagarniss/praktikum-rsi

1. Setup dan Migrasi Database

Langkah pertama dalam pengerjaan tugas ini adalah melakukan inisialisasi *backend* menggunakan Alembic untuk menghubungkan aplikasi FastAPI dengan sistem manajemen basis data (DBMS). Setelah koneksi berhasil dikonfigurasi, skema *database* diterjemahkan ke dalam bentuk tabel melalui proses migrasi.

```
PS D:\praktikum3-rsi\fullstack-app\backend> poetry run alembic upgrade head
INFO  [alembic.runtime.migration] Context impl PostgresqlImpl.
INFO  [alembic.runtime.migration] Will assume transactional DDL.
INFO  [alembic.runtime.migration] Running upgrade  -> 2d02209263e9, create all tables
```

Gambar 1.1 Proses penerapan skrip migrasi ke basis data menggunakan perintah alembic upgrade head



Gambar 1.2 Antarmuka DBeaver yang menampilkan tabel-tabel hasil migrasi pada basis data acara-rsi

2. Dokumentasi Endpoint API (Swagger)

Setelah skema basis data terbentuk, implementasi operasi CRUD (*Create, Read, Update, Delete*) dilakukan menggunakan arsitektur *Controller, Service,* dan *Repository*. Seluruh *endpoint* API kemudian diuji menggunakan antarmuka interaktif Swagger UI. Berikut adalah hasil pengujian dengan skenario berhasil (*Success*) dan gagal (*Failed*).

2.1. Entity: ROLE

Roles

GET	/roles/	Get All Roles Route
POST	/roles/	Create Role Route
GET	/roles/{role_id}	Get Role By Id Route
PUT	/roles/{role_id}	Update Role Route
DELETE	/roles/{role_id}	Delete Role Route

Gambar 2.1 Tampilan Antarmuka Swagger UI untuk Endpoint *Role.*

A. POST /roles (Create Role)

- **Success**

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/roles/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Admin"
}'
```

Request URL

```
http://127.0.0.1:8000/roles/
```

Server response

Code	Details
201	Response body

```
{
  "id": 1,
  "name": "Admin"
}
```

Download

Response headers

```
content-length: 23
content-type: application/json
date: Thu,09 Apr 2026 06:12:09 GMT
server: uvicorn
```

Gambar 2.2 Respons sukses saat menambahkan data *Role* baru.

- **Failed**

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/roles/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": " "
}'
```

Request URL

```
http://127.0.0.1:8000/roles/
```

Server response

Code	Details

400 *Undocumented*	Error: Bad Request
	Response body
	`{ "detail": "Nama role gak boleh kosong!" }` Download
	Response headers
	`content-length: 40`
	`content-type: application/json`
	`date: Thu,09 Apr 2026 06:45:59 GMT`
	`server: uvicorn`

Gambar 2.3 Respons gagal saat menambahkan data *Role* kosong.

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/roles/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Admin Kayang"
}'
```

Request URL

```
http://127.0.0.1:8000/roles/
```

Server response

Code	Details

400 *Undocumented*	Error: Bad Request
	Response body
	`{ "detail": "Role 'Admin Kayang' sudah ada!" }` Download
	Response headers
	`content-length: 43`
	`content-type: application/json`
	`date: Thu,09 Apr 2026 06:48:02 GMT`
	`server: uvicorn`

Gambar 2.4 Respons gagal saat menambahkan data *Role* yang sudah ada.

B. GET /roles (Get All / Get by ID)

- **Success**

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/roles/' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/roles/
```

Server response

Code	Details
200	Response body

```
[
  {
    "id": 2,
    "name": "User Kadipiro"
  },
  {
    "id": 1,
    "name": "Admin Kayang"
  }
]
```

Download

Response headers

```
content-length: 64
content-type: application/json
date: Thu,09 Apr 2026 06:32:04 GMT
server: uvicorn
```

Gambar 2.5 Respons sukses saat mengambil semua data *Role*.

C. GET /roles/{id} (Get Role by ID)

- **Success**

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/roles/1' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/roles/1
```

Server response

Code	Details
200	Response body

```
{
  "id": 1,
  "name": "Admin Kayang"
}
```

Download

Response headers

```
content-length: 30
content-type: application/json
date: Thu,09 Apr 2026 07:02:48 GMT
server: uvicorn
```

Gambar 2.6 Respons sukses saat mengambil data *Role*.

- **Failed**

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/roles/9999' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/roles/9999
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

```
{
  "detail": "Role dengan ID 9999 tidak ditemukan!"
}
```

Download

Response headers

```
content-length: 49
content-type: application/json
date: Thu,09 Apr 2026 06:52:35 GMT
server: uvicorn
```

Gambar 2.7 Respons gagal saat mengambil data *Role.*

D. PUT / PATCH /roles/{id} (Update Role)

- **Success**

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/roles/1' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Admin Kayang"
}'
```

Request URL

```
http://127.0.0.1:8000/roles/1
```

Server response

Code	Details
200	

Response body

```
{
  "id": 1,
  "name": "Admin Kayang"
}
```

Download

Response headers

```
content-length: 30
content-type: application/json
date: Thu,09 Apr 2026 06:21:45 GMT
server: uvicorn
```

Gambar 2.8 Respons sukses saat memperbarui data *Role.*

- **Failed**

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/roles/9999' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Admin Kayang"
}'
```

Request URL

```
http://127.0.0.1:8000/roles/9999
```

Server response

Code	Details

404 *Undocumented*	Error: Not Found
	Response body
	`{ "detail": "Role dengan ID 9999 tidak ditemukan!" }` Download
	Response headers
	`content-length: 49`
	`content-type: application/json`
	`date: Thu,09 Apr 2026 07:01:21 GMT`
	`server: uvicorn`

Gambar 2.9 Respons gagal saat memperbarui data *Role*.

E. DELETE /roles/{id} (Delete Role)

- **Success**

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/roles/2' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/roles/2
```

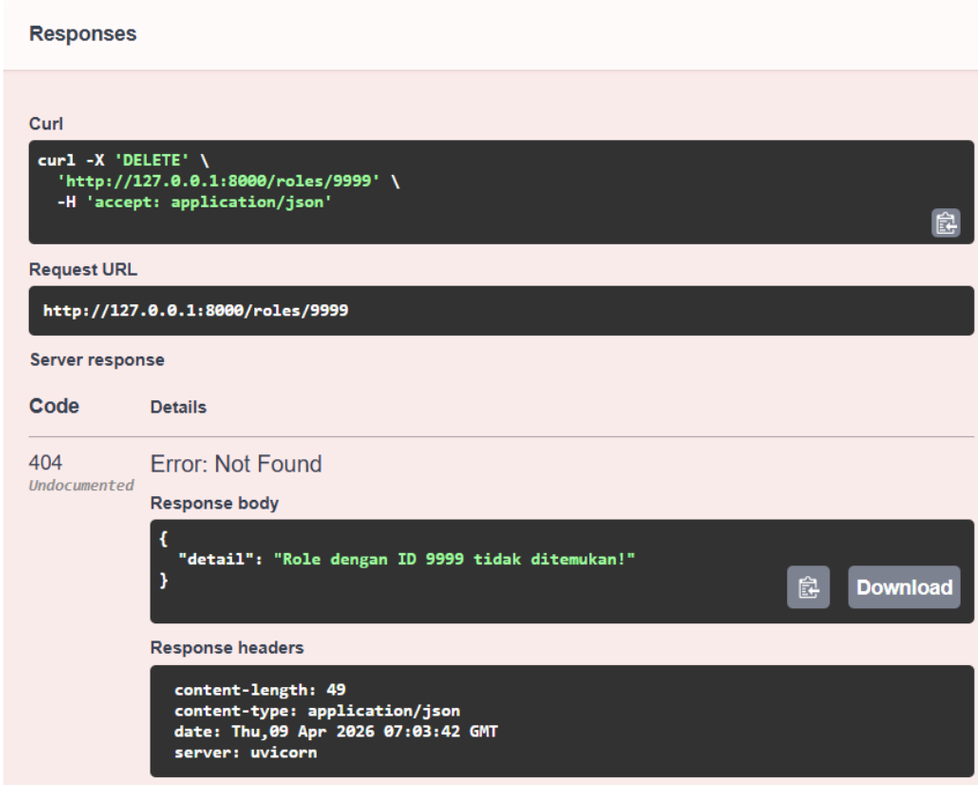
Server response

Code	Details

200	Response body
	`{ "message": "Role dengan ID 2 berhasil dihapus" }` Download
	Response headers
	`content-length: 47`
	`content-type: application/json`
	`date: Thu,09 Apr 2026 06:42:34 GMT`
	`server: uvicorn`

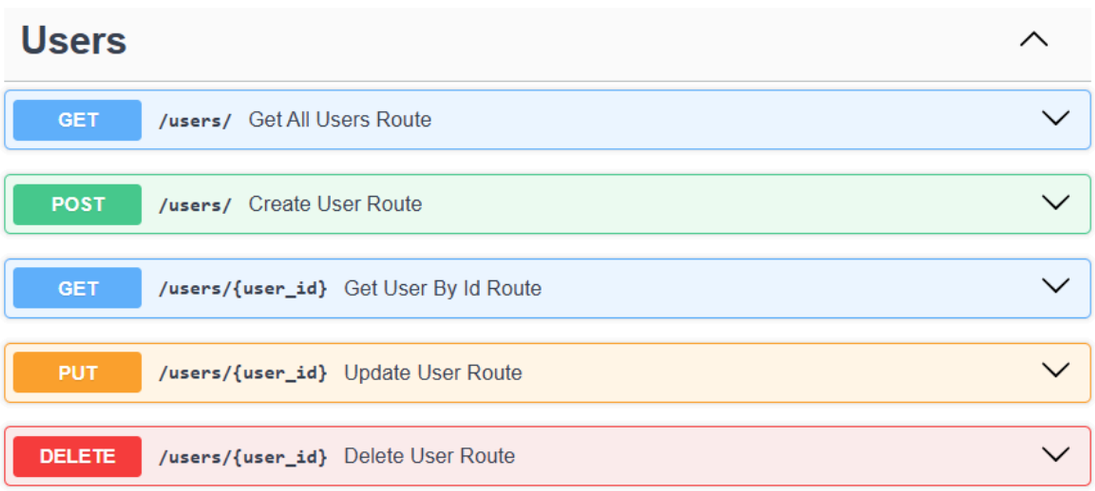
Gambar 2.10 Respons sukses saat menghapus data *Role*.

- **Failed**



Gambar 2.11 Respons gagal saat menghapus data *Role*.

2.2. Entity: USER



Gambar 2.12 Tampilan Antarmuka Swagger UI untuk Endpoint *User*.

A. POST /users (Create User)

- **Success**

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/users/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "first_name": "kenny",
  "last_name": "mutohar",
  "whatsapp": "12345"
}'
```

Request URL

```
http://127.0.0.1:8000/users/
```

Server response

Code	Details
201	Response body

```
{
  "id": 2,
  "first_name": "kenny",
  "last_name": "mutohar",
  "whatsapp": "12345",
  "created_at": "2026-04-09T23:09:53.385911",
  "updated_at": "2026-04-09T23:09:53.385911"
}
```

Gambar 2.13 Respons sukses saat menambahkan data *User* baru.

- **Failed**

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/users/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "first_name": "kenny"
  "last_name": "mutohar",
  "whatsapp": "12345"
}'
```

Request URL

```
http://127.0.0.1:8000/users/
```

Server response

Code	Details
422	Error: Unprocessable Entity

Response body

```
{
  "detail": [
    {
      "type": "json_invalid",
      "loc": [
        "body",
        28
      ],
      "msg": "JSON decode error",
      "input": {},
      "ctx": {
        "error": "Expecting ',' delimiter"
      }
    }
  ]
}
```

Gambar 2.14 Respons gagal saat menambahkan data *User*.

B. GET /users (Get All / Get by ID)

- **Success**

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/users/2' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/users/2
```

Server response

Code	Details
200	Response body

```
{
  "id": 2,
  "first_name": "kenny",
  "last_name": "mutohar",
  "whatsapp": "12345",
  "created_at": "2026-04-09T23:09:53.385911",
  "updated_at": "2026-04-09T23:09:53.385911"
}
```

Gambar 2.15 Respons sukses saat mengambil data *User*.

- **Failed**

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/users/5' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/users/5
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

```
{
  "detail": "User dengan id 5 tidak ditemukan"
}
```

Gambar 2.16 Respons gagal saat mengambil data *User*.

C. PUT / PATCH /users/{id} (Update User)

- **Success**

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/users/2' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "first_name": "kenny",
  "last_name": "kumar",
  "whatsapp": "6789"
}'
```

Request URL

```
http://127.0.0.1:8000/users/2
```

Server response

Code	Details
200	Response body

```
{
  "id": 2,
  "first_name": "kenny",
  "last_name": "kumar",
  "whatsapp": "6789",
  "created_at": "2026-04-09T23:09:53.385911",
  "updated_at": "2026-04-09T23:17:03.896007"
}
```

Gambar 2.17 Respons sukses saat memperbarui data *User*.

- **Failed**

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/users/8' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "first_name": "kenny",
  "last_name": "kumar",
  "whatsapp": "6789"
}'
```

Request URL

```
http://127.0.0.1:8000/users/8
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

```
{
  "detail": "User dengan id 8 tidak ditemukan"
}
```

Gambar 2.18 Respons gagal saat memperbarui data *User*.

D. DELETE /users/{id} (Delete User)

- **Success**

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/users/2' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/users/2
```

Server response

Code	Details
200	Response body

```
{
  "id": 2,
  "first_name": "kenny",
  "last_name": "kumar",
  "whatsapp": "6789",
  "created_at": "2026-04-09T23:09:53.385911",
  "updated_at": "2026-04-09T23:17:03.896007"
}
```

Gambar 2.19 Respons sukses saat menghapus data *User*.

- **Failed**

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/users/8' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/users/8
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

```
{
  "detail": "User dengan id <built-in function id> tidak ditemukan"
}
```

Gambar 2.20 Respons gagal saat menghapus data *User*.

2.3. Entity: ACCOUNT

Accounts

GET	/accounts/	Get All Accounts Route
POST	/accounts/	Create Account Route
GET	/accounts/{account_id}	Get Account By Id Route
PUT	/accounts/{account_id}	Update Account Route
DELETE	/accounts/{account_id}	Delete Account Route

Gambar 2.21 Tampilan Antarmuka Swagger UI untuk Endpoint *Account.*

A. POST /accounts (Create Account)

- **Success**

```
curl -X 'POST' \
  'http://127.0.0.1:8000/accounts/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": 1,
  "role_id": 1,
  "email": "tatang@example.com",
  "username": "tangtung",
  "password": "kudus"
}'
```

Request URL

```
http://127.0.0.1:8000/accounts/
```

Server response

Code	Details
201	Response body

```
{
  "id": 6,
  "user_id": 1,
  "role_id": 1,
  "email": "tatang@example.com",
  "username": "tangtung",
  "created_at": "2026-04-09T22:54:29.497081",
  "updated_at": "2026-04-09T22:54:29.497081"
}
```

Response headers

```
content-length: 167
content-type: application/json
date: Thu,09 Apr 2026 15:54:28 GMT
server: uvicorn
```

Gambar 2.22 Respons sukses saat menambahkan data *Account* baru.

- **Failed**

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/accounts/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": 2,
  "role_id": 2,
  "email": "tatang@example.com",
  "username": "tangtung",
  "password": "kudus"
}'
```

Request URL

```
http://127.0.0.1:8000/accounts/
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

```
{
  "detail": "User tidak ditemukan"
}
```

Response headers

```
content-length: 33
content-type: application/json
date: Thu,09 Apr 2026 15:55:40 GMT
server: uvicorn
```

Gambar 2.23 Respons gagal saat menambahkan data *Account*.

B. GET /accounts (Get All / Get by ID)

- **Success**

200

Response body

```
{
  "id": 6,
  "user_id": 1,
  "role_id": 1,
  "email": "tatang@example.com",
  "username": "tangtung",
  "created_at": "2026-04-09T22:54:29.497081",
  "updated_at": "2026-04-09T22:54:29.497081"
}
```

Gambar 2.24 Respons sukses saat mengambil data *Account*.

- **Failed**

404
Undocumented

Error: Not Found

Response body

```
{
  "detail": "Akun dengan ID 7 tidak ditemukan"
}
```

Response headers

```
content-length: 45
content-type: application/json
date: Thu,09 Apr 2026 15:58:56 GMT
server: uvicorn
```

Gambar 2.25 Respons gagal saat mengambil data *Account.*

C. PUT / PATCH /accounts/{id} (Update Account)

- **Success**

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/accounts/9' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": 1,
  "role_id": 1,
  "email": "titing@example.com",
  "username": "mutiara",
  "password": "malika"
}'
```

Request URL

```
http://127.0.0.1:8000/accounts/9
```

Server response

Code	Details
200	Response body

```
{
  "id": 9,
  "user_id": 1,
  "role_id": 1,
  "email": "titing@example.com",
  "username": "mutiara",
  "created_at": "2026-04-09T23:25:33.330574",
  "updated_at": "2026-04-09T23:25:33.330574"
}
```

Gambar 2.26 Respons sukses saat memperbarui data *Account*.

- **Failed**

404
Undocumented

Error: Not Found

Response body

```
{
  "detail": "Akun dengan ID 7 tidak ditemukan"
}
```

Response headers

```
content-length: 45
content-type: application/json
date: Thu,09 Apr 2026 16:02:16 GMT
server: uvicorn
```

Gambar 2.27 Respons gagal saat memperbarui data *Account*.

D. DELETE /accounts/{id} (Delete Account)

- **Success**

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/accounts/6' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/accounts/6
```

Server response

Code	Details
200	**Response body**

```
{
  "message": "Akun dengan ID 6 berhasil dihapus"
}
```

Response headers

```
content-length: 47
content-type: application/json
date: Thu,09 Apr 2026 16:03:17 GMT
server: uvicorn
```

Gambar 2.28 Respons sukses saat menghapus data *Account*.

- **Failed**

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/accounts/7' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/accounts/7
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

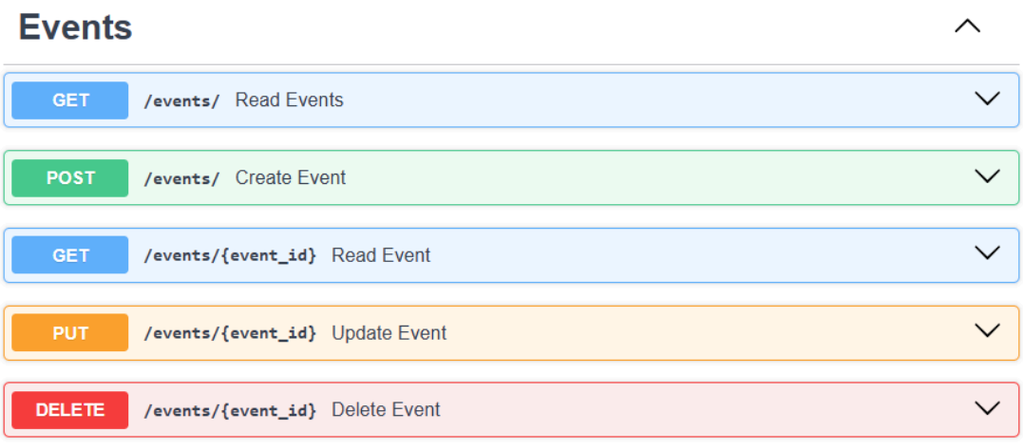
```
{
  "detail": "Akun dengan ID 7 tidak ditemukan"
}
```

Response headers

```
content-length: 45
content-type: application/json
date: Thu,09 Apr 2026 16:07:02 GMT
server: uvicorn
```

Gambar 2.29 Respons gagal saat menghapus data *Account.*

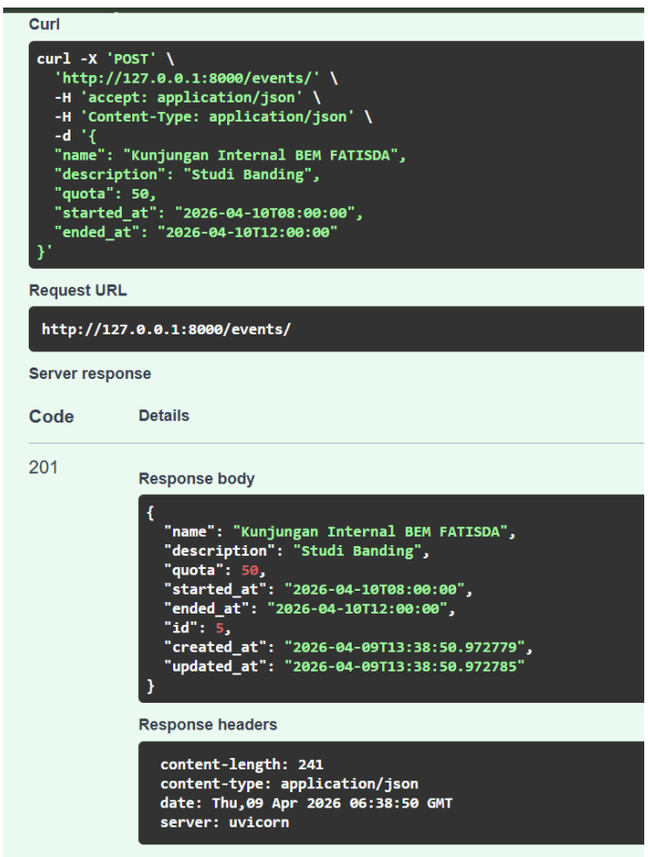
2.4. Entity: EVENT



Gambar 2.30 Tampilan Antarmuka Swagger UI untuk Endpoint *Event.*

A. POST /events (Create Event)

- **Success**



Gambar 2.31 Respons sukses saat menambahkan data *Event* baru.

- **Failed**

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/events/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": " ",
  "description": "Studi Banding",
  "quota": 50,
  "started_at": "2026-04-10T08:00:00",
  "ended_at": "2026-04-10T12:00:00"
}'
```

Request URL

```
http://127.0.0.1:8000/events/
```

Server response

Code	Details
400 *Undocumented*	Error: Bad Request

Response body

```
{
  "detail": "Nama event wajib diisi!"
}
```

Response headers

```
content-length: 36
content-type: application/json
date: Thu,09 Apr 2026 06:39:41 GMT
server: uvicorn
```

Gambar 2.32 Respons gagal saat menambahkan data *Event*.

B. GET /events (Get All / Get by ID)

- **Success**

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/events/5' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/events/5
```

Server response

Code	Details
200	

Response body

```
{
  "name": "Kunjungan Internal BEM FATISDA",
  "description": "Studi Banding",
  "quota": 50,
  "started_at": "2026-04-10T08:00:00",
  "ended_at": "2026-04-10T12:00:00",
  "id": 5,
  "created_at": "2026-04-09T13:38:50.972779",
  "updated_at": "2026-04-09T13:38:50.972785"
}
```

Response headers

Gambar 2.33 Respons sukses saat mengambil data *Event*.

- **Failed**

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/events/1' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/events/1
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found **Response body** `{ "detail": "Event tidak ditemukan" }`

Gambar 2.34 Respons gagal saat mengambil data *Event*.

C. PUT / PATCH /events/{id} (Update Event)

- **Success**

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/events/5' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Kunjungan Internal BEM FATISDA x BEM SV",
  "description": "Studi Banding dan Kolaborasi Proker",
  "quota": 100,
  "started_at": "2026-04-10T08:00:00",
  "ended_at": "2026-04-10T12:00:00"
}'
```

Request URL

```
http://127.0.0.1:8000/events/5
```

Server response

Code	Details

200	Response body
	`{ "name": "Kunjungan Internal BEM FATISDA x BEM SV", "description": "Studi Banding dan Kolaborasi Proker", "quota": 100, "started_at": "2026-04-10T08:00:00", "ended_at": "2026-04-10T12:00:00", "id": 5, "created_at": "2026-04-09T13:38:50.972779", "updated_at": "2026-04-09T13:47:02.179002" }`
	Response headers
	`content-length: 273`
	`content-type: application/json`
	`date: Thu,09 Apr 2026 06:47:01 GMT`
	`server: uvicorn`

Gambar 2.35 Respons sukses saat memperbarui data *Event*.

- **Failed**

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/events/4' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "name": "Kunjungan Internal BEM FATISDA x BEM SV",
  "description": "Studi Banding dan Kolaborasi Proker",
  "quota": 100,
  "started_at": "2026-04-10T08:00:00",
  "ended_at": "2026-04-10T12:00:00"
}'
```

Request URL

```
http://127.0.0.1:8000/events/4
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

```
{
  "detail": "Event tidak ditemukan"
}
```

Response headers

```
content-length: 34
content-type: application/json
date: Thu,09 Apr 2026 06:47:40 GMT
server: uvicorn
```

Gambar 2.36 Respons gagal saat memperbarui data *Event*.

D. DELETE /events/{id} (Delete Event)

- **Success**

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/events/1' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/events/1
```

Server response

Code	Details
200	

Response body

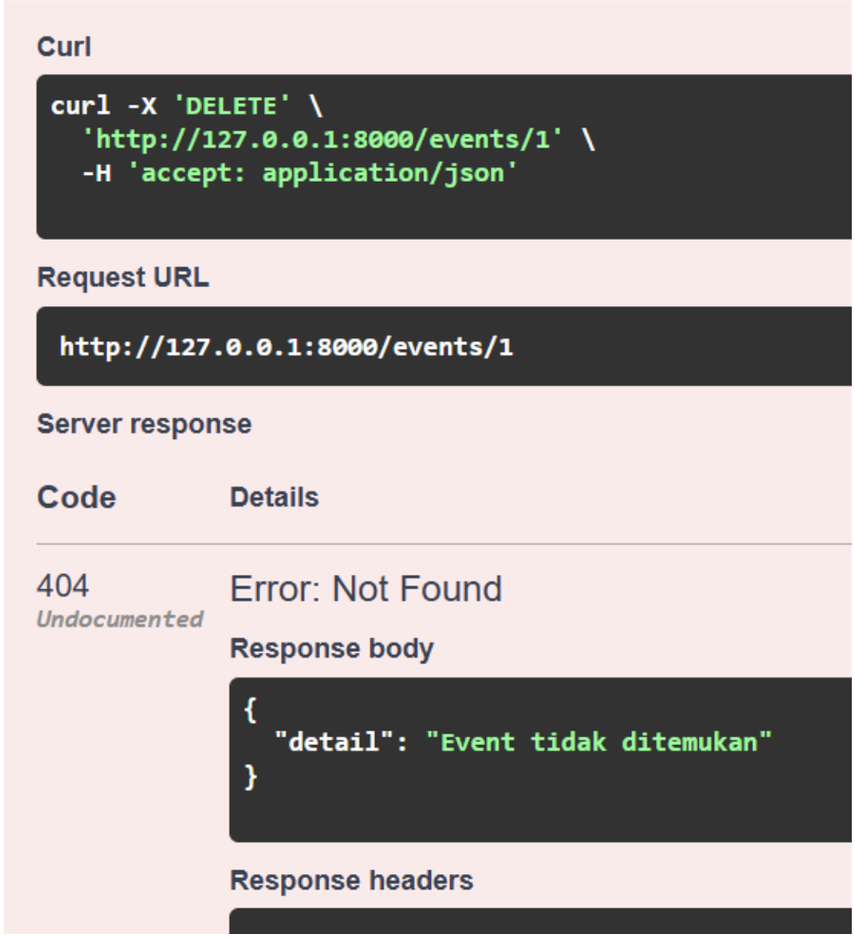
```
{
  "message": "Event dengan ID 1 berhasil dihapus"
}
```

Response headers

```
content-length: 48
content-type: application/json
```

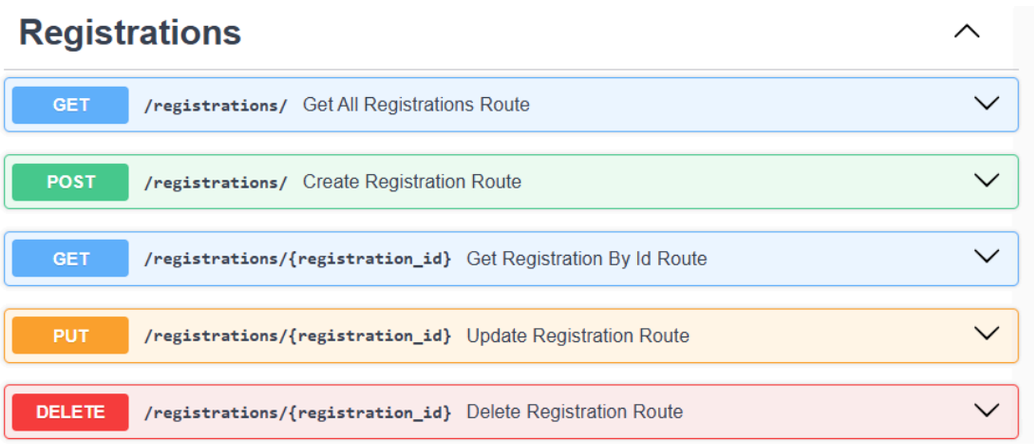
Gambar 2.37 Respons sukses saat menghapus data *Event*.

- **Failed**



Gambar 2.38 Respons gagal saat menghapus data *Event.*

2.5. Entity: REGISTRATION



Gambar 2.39 Tampilan Antarmuka Swagger UI untuk Endpoint *Registration.*

A. POST /registrations (Create Registration)

- **Success**

Curl

```
curl -X 'POST' \
  'http://localhost:8000/registrations/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": 1,
  "event_id": 1
}'
```

Request URL

```
http://localhost:8000/registrations/
```

Server response

Code	Details

201	Response body
	`{ "id": 4 }`
	Response headers
	`content-length: 8`
	`content-type: application/json`
	`date: Fri,10 Apr 2026 01:21:24 GMT`
	`server: uvicorn`

Responses

Gambar 2.40 Respons sukses saat menambahkan data *Registration* baru.

- **Failed**

Curl

```
curl -X 'POST' \
  'http://localhost:8000/registrations/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": 1,
  "event_id": 1
}'
```

Request URL

```
http://localhost:8000/registrations/
```

Server response

Code	Details

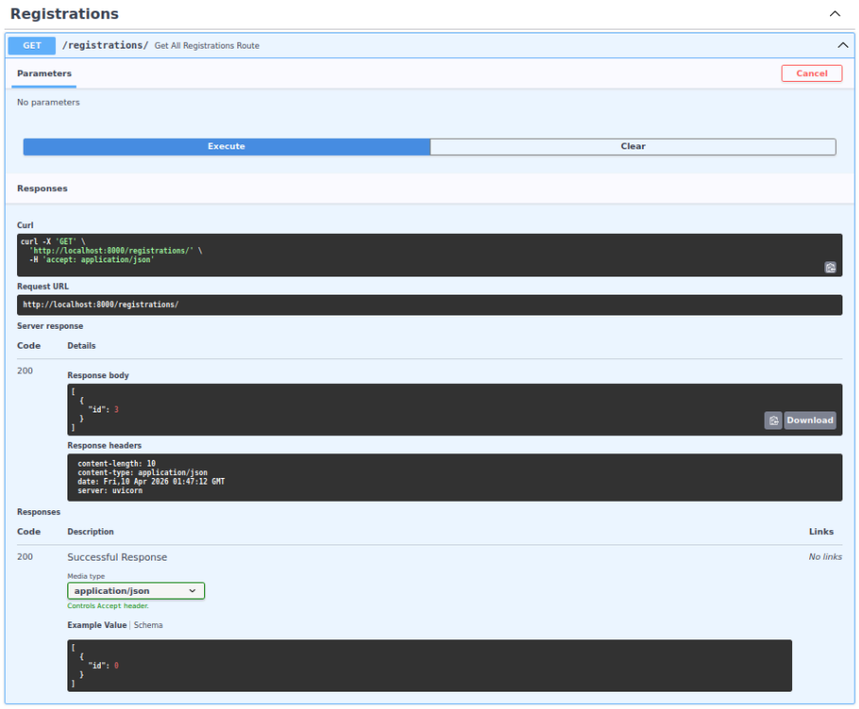
400
Undocumented
	Error: Bad Request
	Response body
	`{ "detail": "User is already registered for this event" }`
	Response headers
	`content-length: 54`
	`content-type: application/json`
	`date: Fri,10 Apr 2026 01:51:03 GMT`
	`server: uvicorn`

Responses

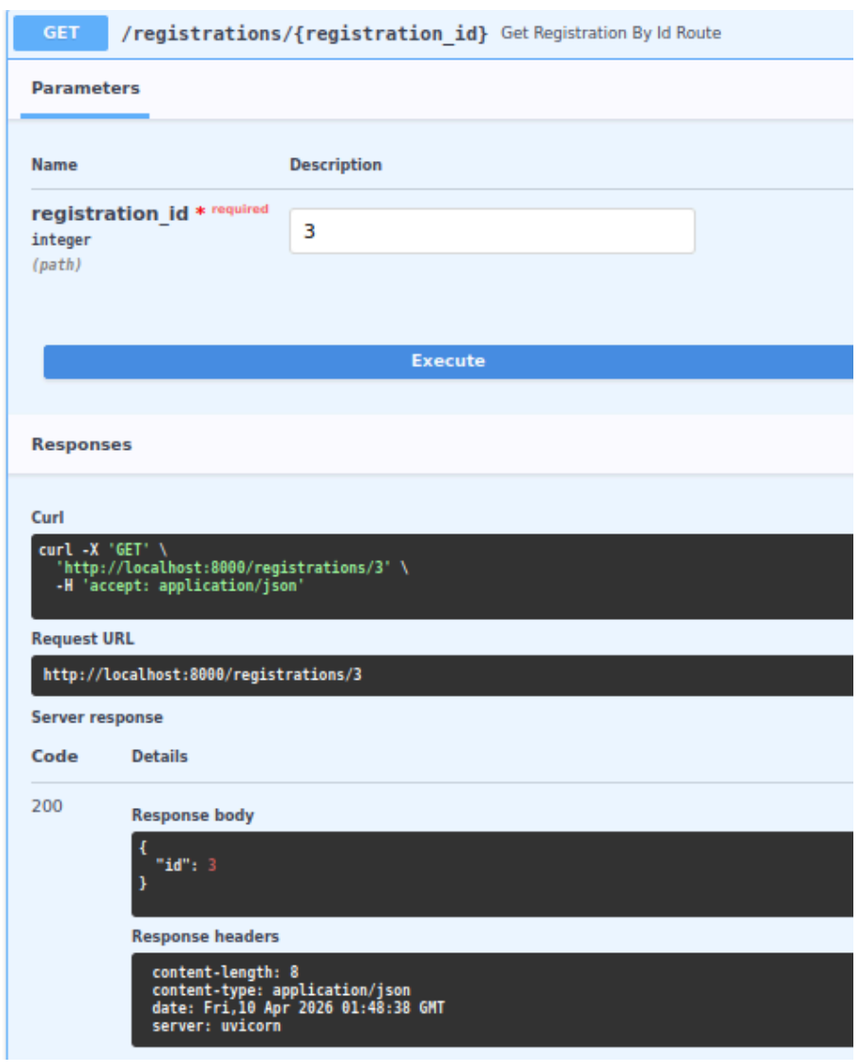
Gambar 2.41 Respons gagal saat menambahkan data *Registration*.

B. GET /registrations (Get All / Get by ID)

- **Success**

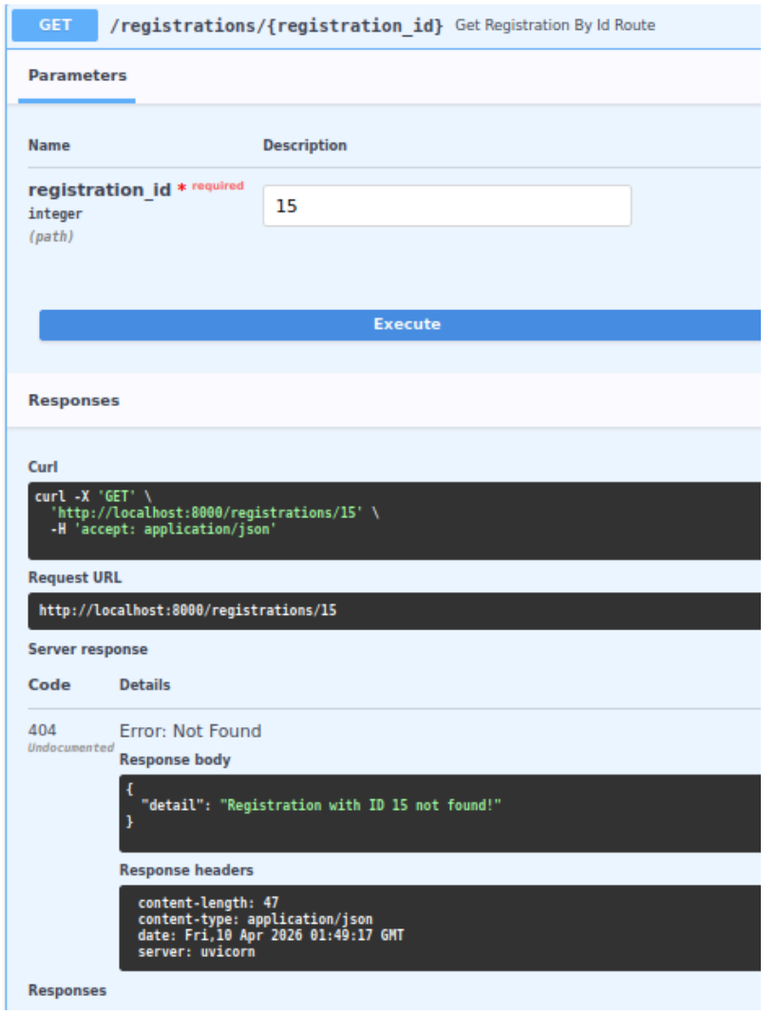


Gambar 2.42 Respons sukses saat mengambil semua data *Registrations*.



Gambar 2.43 Respons sukses saat mengambil data *Registrations* by id.

- **Failed**



Gambar 2.44 Respons gagal saat mengambil data *Registration*.

C. PUT / PATCH /registrations/{id} (Update Registration)

- **Success**

Gambar 2.45 Respons sukses saat memperbarui data *Registration*.

- **Failed**

Curl

```
curl -X 'PUT' \
  'http://localhost:8000/registrations/3' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
  "user_id": 1,
  "event_id": 1
}'
```

Request URL

```
http://localhost:8000/registrations/3
```

Server response

Code	Details
400 *Undocumented*	Error: Bad Request

Response body

```
{
  "detail": "User is already registered for this event"
}
```

Response headers

```
content-length: 54
content-type: application/json
date: Fri,10 Apr 2026 01:55:21 GMT
server: uvicorn
```

Responses

Gambar 2.46 Respons gagal saat memperbarui data *Registration*.

D. DELETE /registrations/{id} (Delete Registration)

- **Success**

Curl

```
curl -X 'DELETE' \
  'http://localhost:8000/registrations/5' \
  -H 'accept: application/json'
```

Request URL

```
http://localhost:8000/registrations/5
```

Server response

Code	Details
200	

Response body

```
{
  "message": "Registration with ID 5 successfully deleted"
}
```

Response headers

```
content-length: 57
content-type: application/json
date: Fri,10 Apr 2026 01:55:47 GMT
server: uvicorn
```

Gambar 2.47 Respons sukses saat menghapus data *Registration*.

- **Failed**

Curl

```
curl -X 'DELETE' \
  'http://localhost:8000/registrations/5' \
  -H 'accept: application/json'
```

Request URL

```
http://localhost:8000/registrations/5
```

Server response

Code	Details
404 *Undocumented*	Error: Not Found

Response body

```
{
  "detail": "Registration with ID 5 not found!"
}
```

Response headers

```
content-length: 46
content-type: application/json
date: Fri,10 Apr 2026 01:56:25 GMT
server: uvicorn
```

Gambar 2.48 Respons gagal saat menghapus data *Registration*.

KESIMPULAN

Berdasarkan pengerjaan penugasan Backend CRUD API ini, dapat disimpulkan beberapa poin utama:

1. Keberhasilan Setup & Migrasi: Kelompok kami telah berhasil menginisialisasi basis data dan melakukan migrasi skema tabel (Role, User, Account, Event, Registration, dan Log) menggunakan Alembic. Alat ini sangat mempermudah pelacakan versi (versioning) dari basis data layaknya Git pada source code.
2. Implementasi Arsitektur Terstruktur: Pembuatan operasi CRUD tidak dilakukan secara sembarangan, melainkan mengikuti standar arsitektur perangkat lunak yang baik. Pemisahan layer seperti Controller (untuk routing), Service (untuk business logic), Repository (untuk query database), dan DTO (untuk validasi data) membuat program FastAPI kami menjadi sangat modular, rapi, dan mudah dikembangkan ke depannya.
3. Pengujian Endpoint: Seluruh endpoint API telah diuji fungsionalitasnya menggunakan Swagger UI. Sistem terbukti mampu memproses request dengan benar untuk skenario berhasil (mengembalikan status 2xx) maupun menangani error atau validasi yang tidak sesuai pada skenario gagal (mengembalikan status 4xx).